



How it works?

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able cellular communicators. Although the Motorola DynaTAC arrived in 1984 it wouldn't be until the mid-90s that 'cellphones' became popular across the world. By then, the Nokia 5110 Communicator became a beacon of advancement in technology. In 1999 came Intel's Pentium III series of processors, running on 28 million transistors.

With faster processing possible due to massive number of processing components being fitted into chipsets, the rest of the technology world started growing around processor advancements. Coloured displays and graphic-rendering processing units started becoming commonplace. "This is not going to stop soon," Moore famously stated about his own "law", in 1995. And it didn't.

Stepping Down

The mid-2000s and onwards on saw the advent of multi-core processor into daily life, which trickled down to cellphones later. Advanced Micro Devices (AMD) were the first to showcase a dual-core processor in 2004, although Intel happened to release the first commercially-available dual-core processor in 2005 - the Intel Pentium 840 Extreme Edition. The world's first cellphone to ever run on a multi-core processor was the LG Optimus 2X, powered by the dual-core nVidia Tegra 2. These advancements not only signify technical achievements - they empowered the world to achieve further in terms of software, augmentative realities and cognitive branches. Graphic designs started improving drastically, researchers started using multi-core platforms to synthesize more data, advancing medical science, space research and immensely powerful setups capable of powering multiple facets of life.

It is in this phase that the Moore's Law started being threatened. Processors have now reached a stage where adding more cores and components do not drastically improve performance because our usual tasks run their fastest already on existing architectures. Technology has started developing away from the Moore-centric drive. The Moore's Law, in its actual sense, is not a "law". It does not have a prolonged scientific or natural recurring occurrence.

GADGET TIMELINE



Apple 1 (1976)



Apple 2 (1977)



IBM PC (1981)



Apple Macintosh (1984)



Motorola DynaTAC 8000X (1984)



Nokia 5110 (1998)



LG Optimus 2X (2011)

It is more of an observation, that predicted that path of five decades of technological progress. However, silicon chipsets can only be this powerful, and adding more transistors to a processor will not considerably improve performance any further. Additionally, processor clock speeds have come to a standstill. This is because transistor speeds have not really increased much over the last few years, and power consumption optimisation at higher clock speeds cannot be attained. As a result, more data cannot be processed in a single clock cycle. Chipsets have huge lengths of internal wiring, and process impulses are scientifically impossible to be pushed

beyond the speed of light. Light traverses four meters in a nanosecond, and no law can push beyond that.

To look for alternatives to silicon computing, researchers started looking at nanotechnology and molecular computers, completely driving away from Moore's Law. Molecular computing developments involve IBM dabbling with oscillating circuits based on a single carbon nanotube molecule, wherein the entire circuitry depends on a single carbon molecule to speed up processing. This does not rely on the relay of electric impulses through wires to work out processes. Quantum computing tech-